

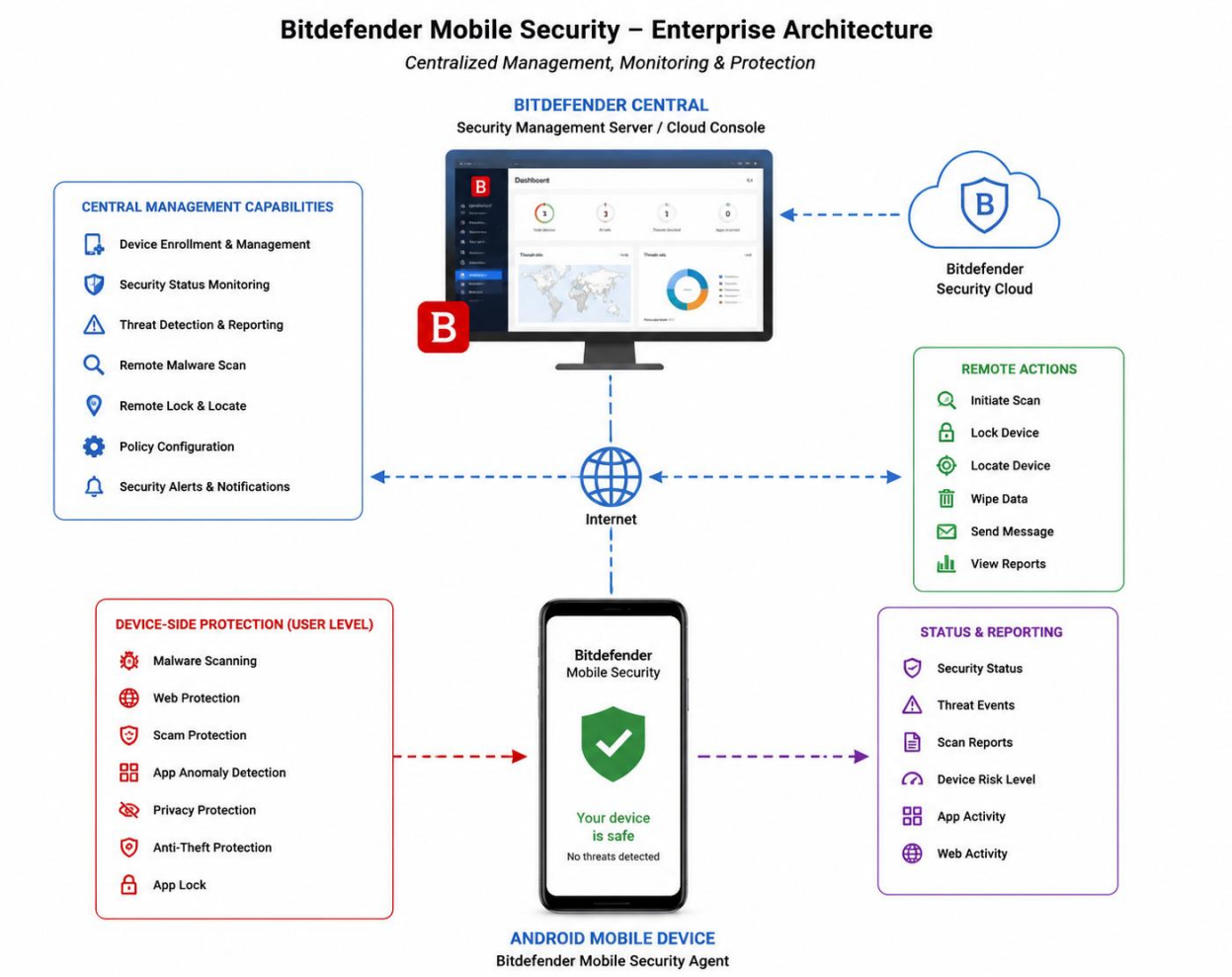
Abdul Wahid

EC-Council CND (Certified Network Defender)

Mobile Device Security – Malware Detection & Enterprise Management (Emsisoft & Bitdefender)

Executive Summary

This module demonstrates the implementation and evaluation of **mobile device security using two different security approaches**. The first part focuses on **standalone mobile malware detection and protection using Emsisoft Mobile Security**, where security scanning and threat identification were performed to analyze suspicious files and ensure device safety. The second part focuses on **Bitdefender Mobile Security with centralized enterprise management**, demonstrating how mobile devices can be protected at the user level while also being monitored and managed through an enterprise security platform. Together, these labs demonstrate how modern mobile security solutions provide malware prevention, threat detection, privacy protection, device monitoring, and centralized security management capabilities.



Network Setup

Device	Quantity	Purpose
Security Management Server	1	Used as a centralized management system for monitoring, reporting, and controlling mobile security activities.
Mobile Device (Android)	1	Used as the protected endpoint for installing security applications, performing malware scans, and testing mobile security features.

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Mobile Malware Detection and Threat Analysis Using Emsisoft Mobile Security

Objective

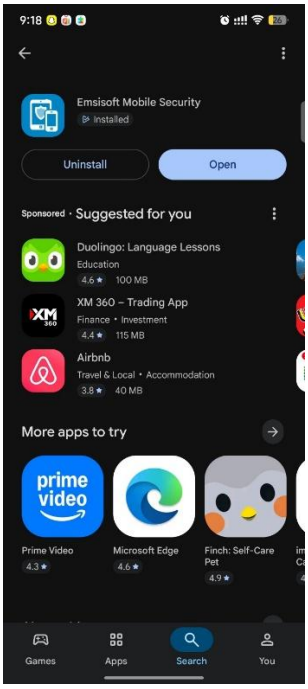
The objective of this lab is to analyze mobile device security using **Emsisoft Mobile Security** by performing malware detection and threat identification. This lab demonstrates how a mobile security solution can scan files, detect malicious content, generate security alerts, and assist users in removing potential threats from their devices.

The lab focuses on understanding:

- Mobile malware detection mechanisms
- Security alerts generated after threat identification
- Analysis of suspicious APK files
- Malware removal and device protection
- The role of antivirus solutions in maintaining mobile device security

Emsisoft Setup

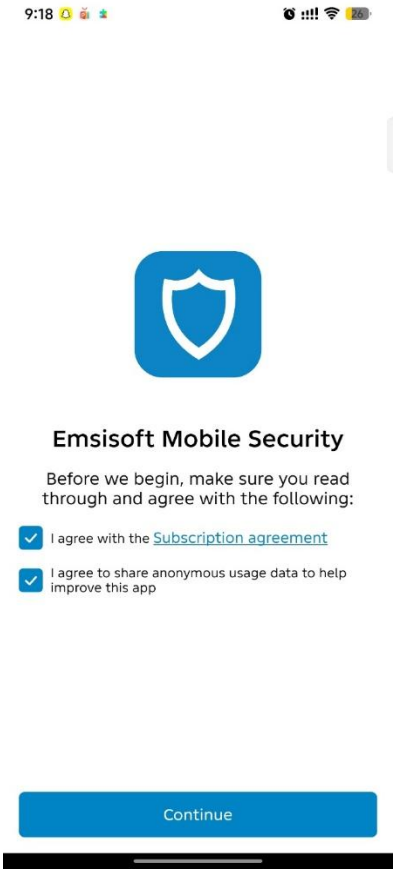
Open play store and install Emsisoft Mobile Security

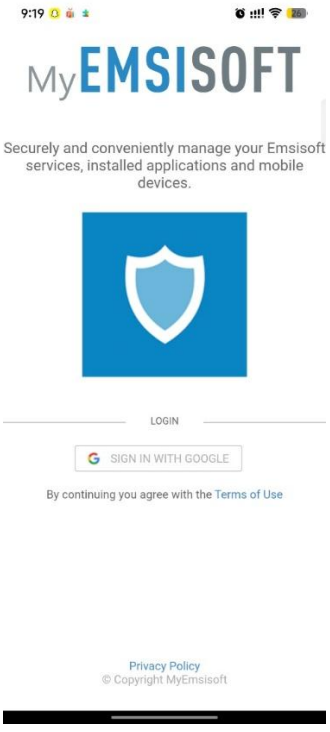


Click on get started



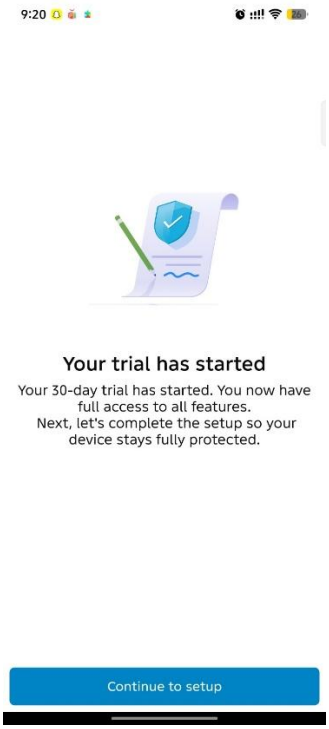
Agree to the terms and conditions





Continuing with a free trial

Click on Continue to Setup

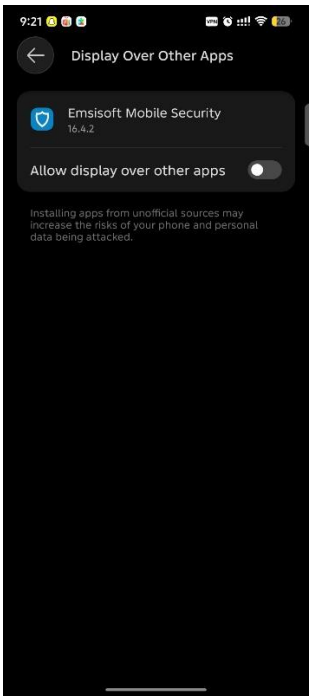


Allow VPN

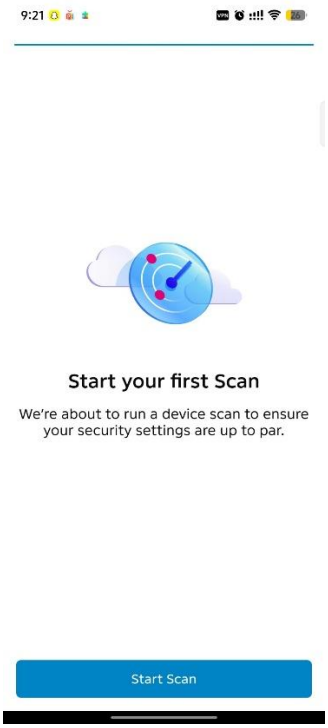


Allow display over other apps

- Turn it on



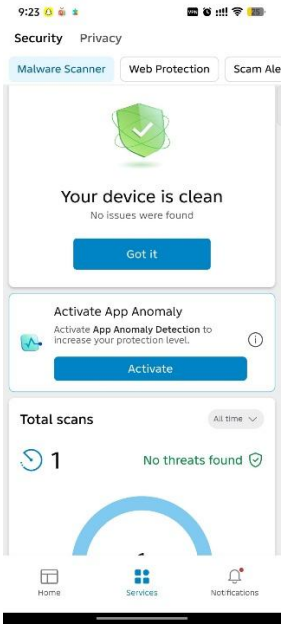
Start your scan



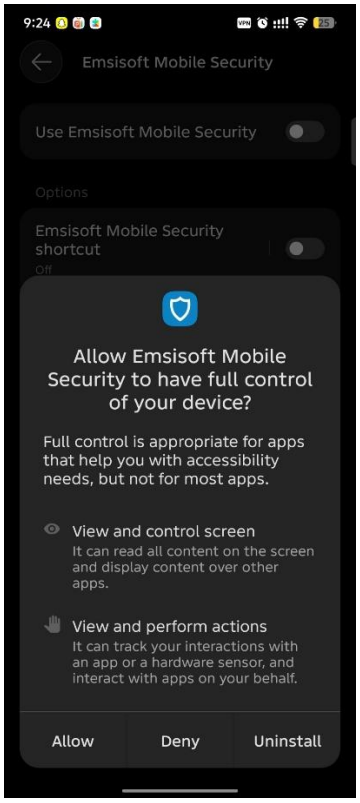
Select Malware Scanner



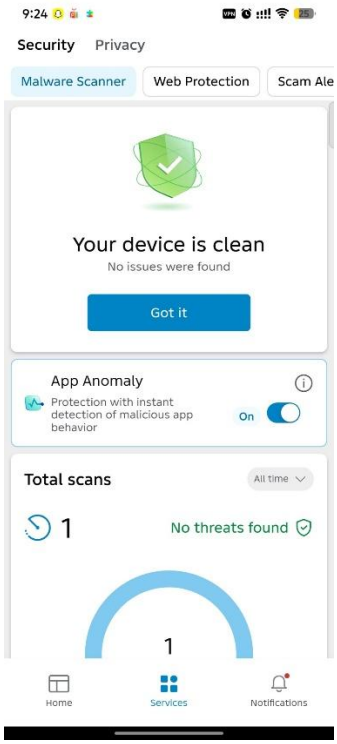
Activate Anomaly App Detection



Click Allow



App Anomaly Detection is ON Successfully

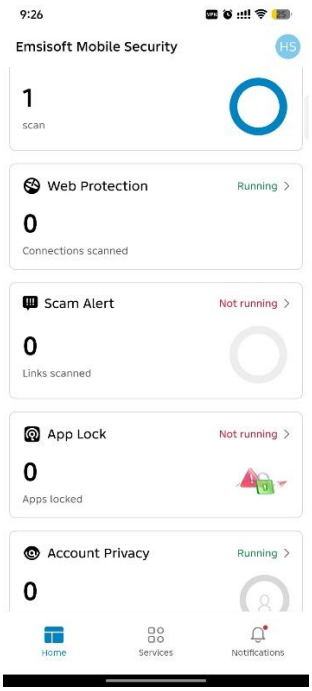


Enable all the Scans in the Malware scanner

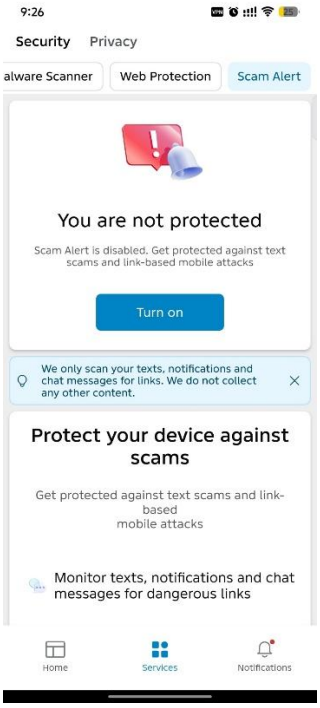


Come back to Home Dashboard

Turn on Scam Alert

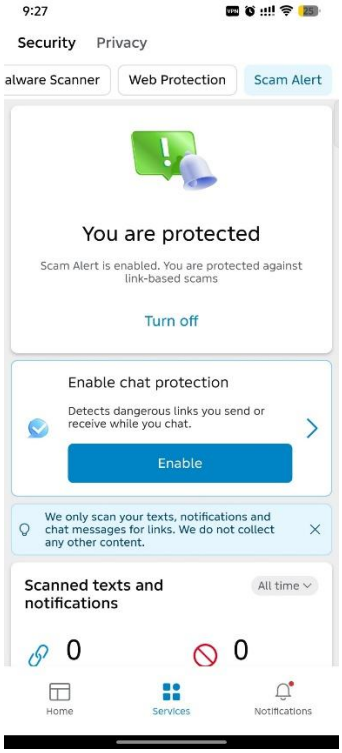


Click on Turn On

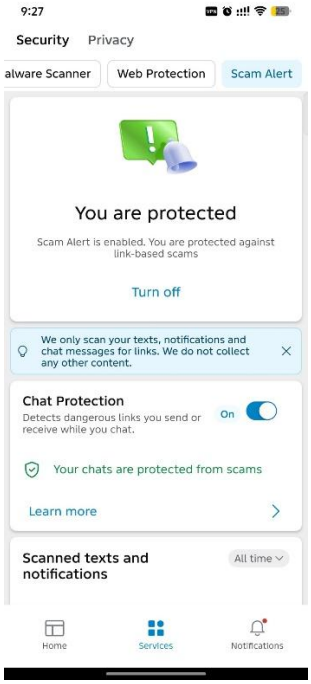


Scam Alert enabled

Click “Enable chat protection”

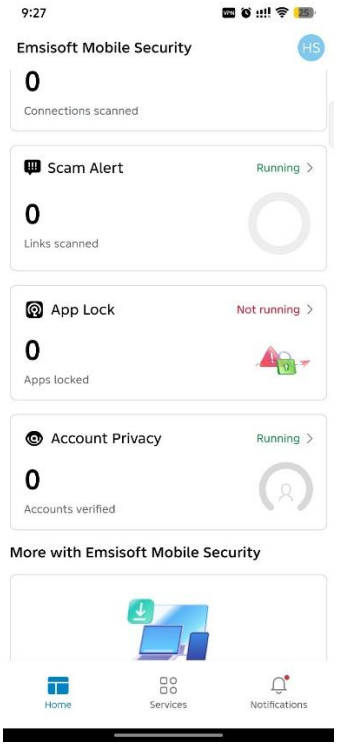


Chat protection enabled



Coming back to Dashboard

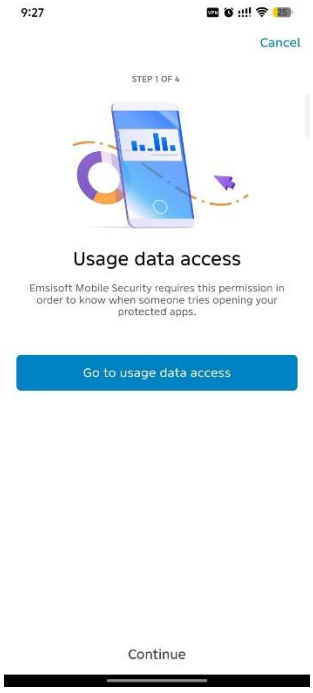
Turn App Lock on



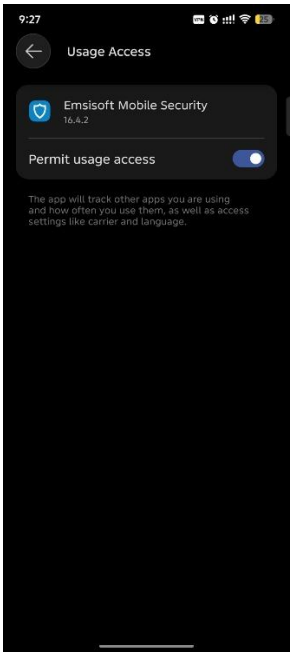
Click on Get Started



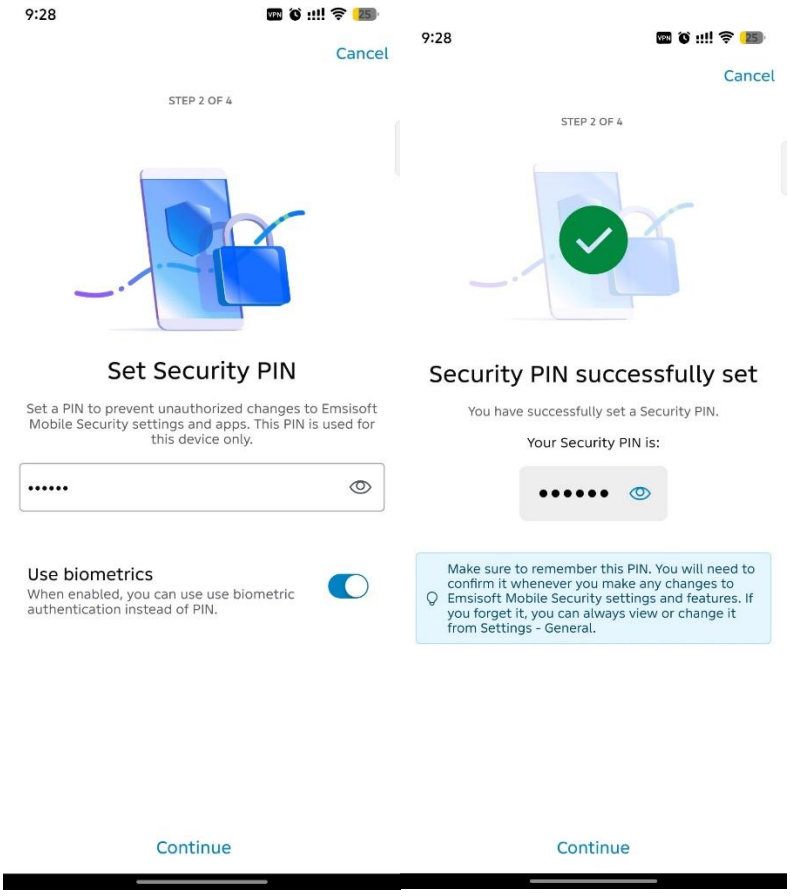
Give usage access to emsisoft



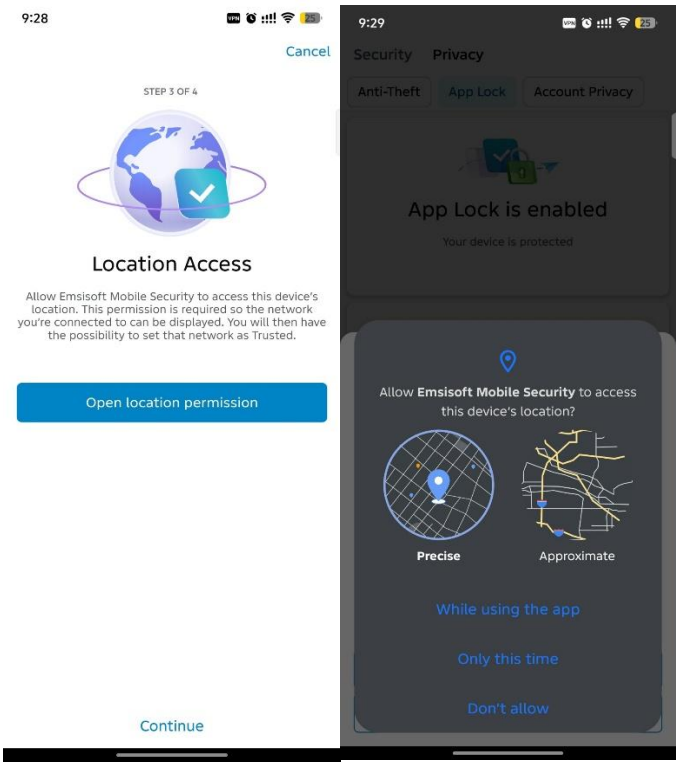
Turn on the Usage access to Emsisoft



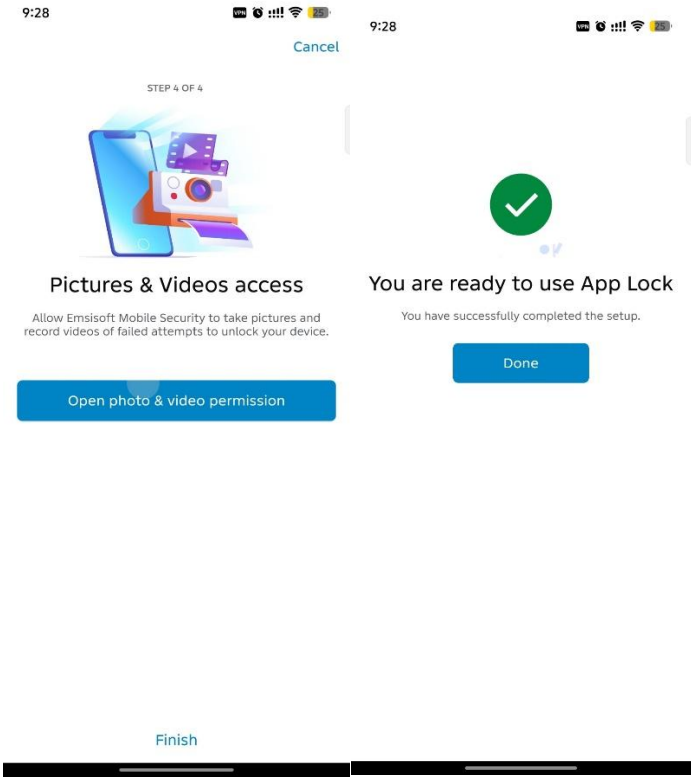
Add a Security Pin



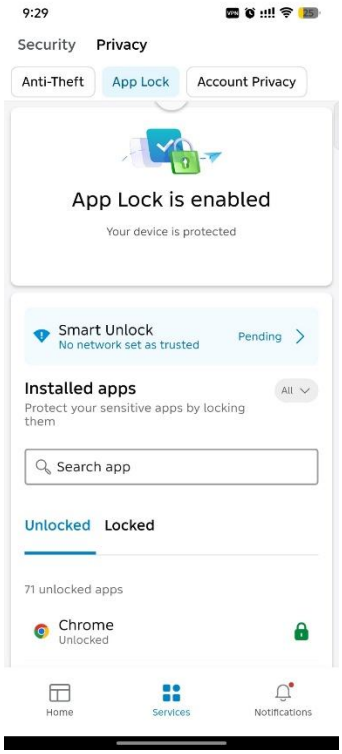
Giving Location Access



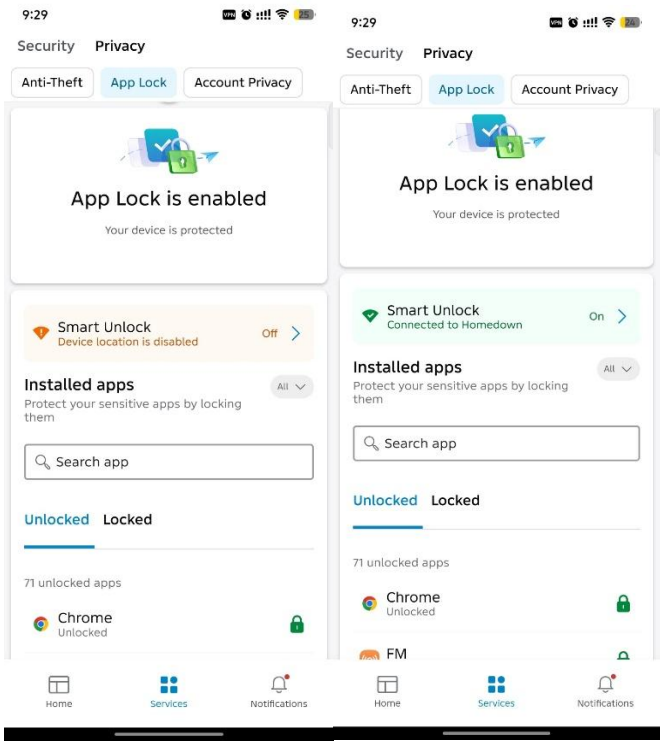
Click Finish & App Lock will be set



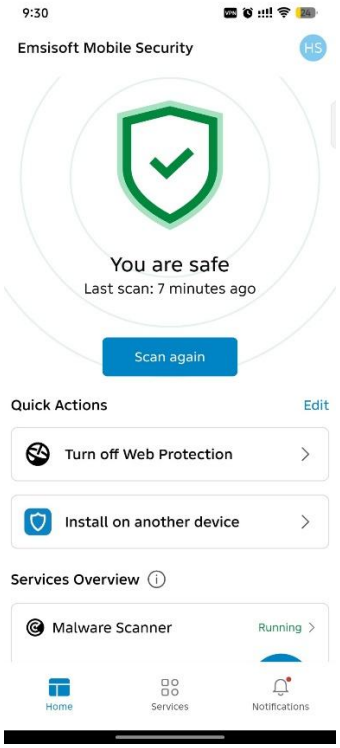
App Lock Enabled Successfully



Turn on Smart Unlock



Emsisoft 100% Ready to do a Scan now

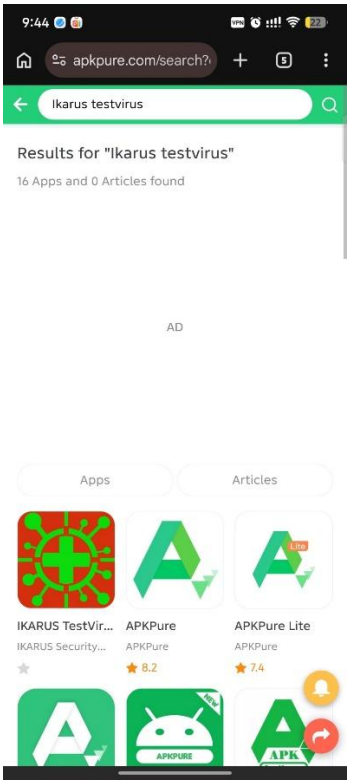


Launching IKARUS for Test

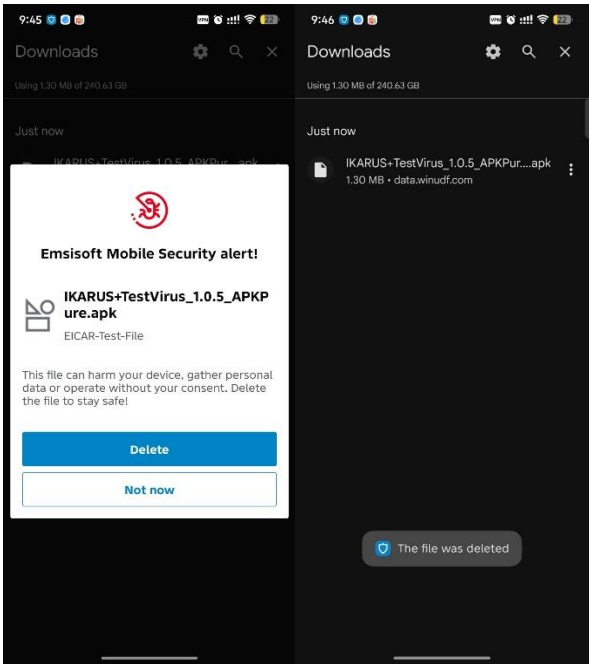
Now let's test Emsisoft by downloading IKARUS in our Mobile

Navigate to apkpure.com

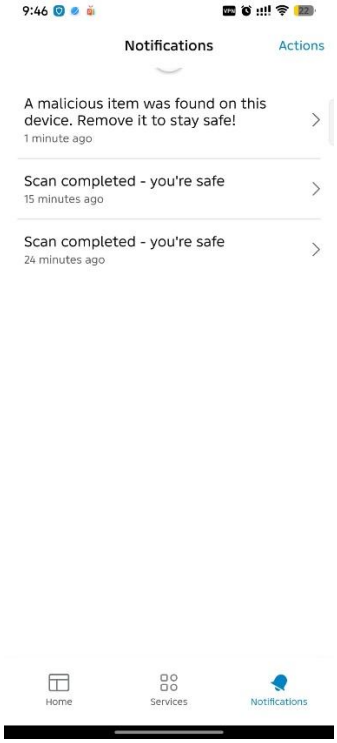
Search & Download the IKARUS



As when the file is downloaded, Emsisoft will immediately detect the downloaded file and we will see an alert & in few seconds you'll see FILE GOT DELETED



Open Notifications in Emsisoft & you'll see what happened with your mobile



Mobile Device Protection and Enterprise Security Management Using Bitdefender Mobile Security

Objective

The objective of this lab is to implement comprehensive mobile security protection using Bitdefender Mobile Security and demonstrate enterprise-level device management through Bitdefender Central.

This lab focuses on two security perspectives:

User-Level Mobile Protection

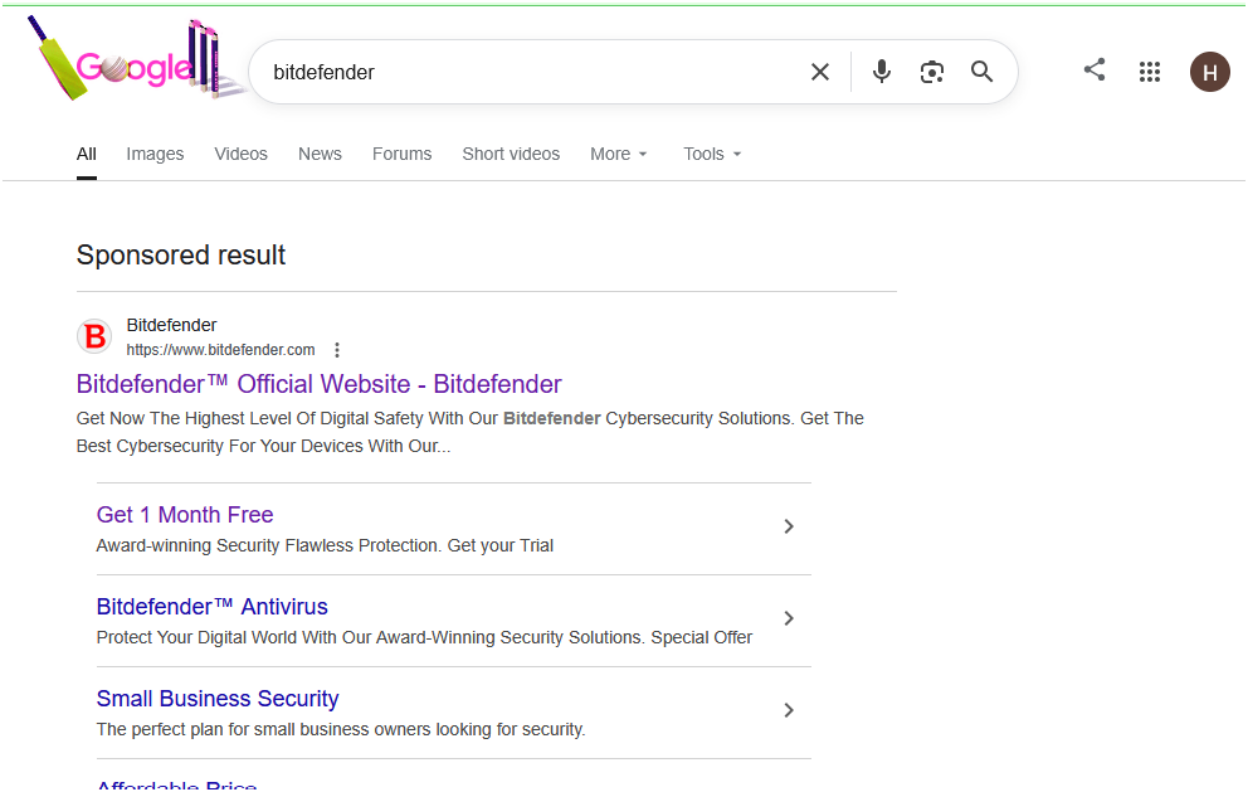
- **Installing and configuring Bitdefender Mobile Security on a personal device**
- **Enabling malware scanning, web protection, scam protection, and application monitoring**
- **Protecting the device against malicious applications, phishing attempts, and unsafe activities**

Enterprise-Level Security Management

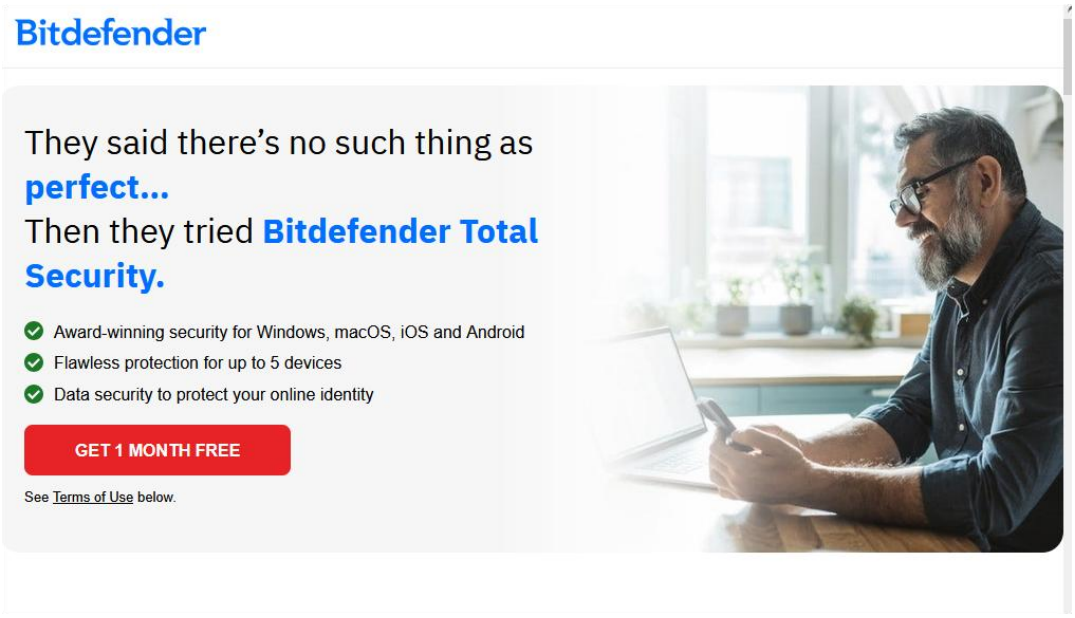
- **Connecting a mobile device with Bitdefender Central**
- **Enrolling and monitoring devices through a centralized security dashboard**
- **Performing remote security actions such as malware scanning and anti-theft management**
- **Monitoring device security status and responding to detected threats**

Bitdefender Setup

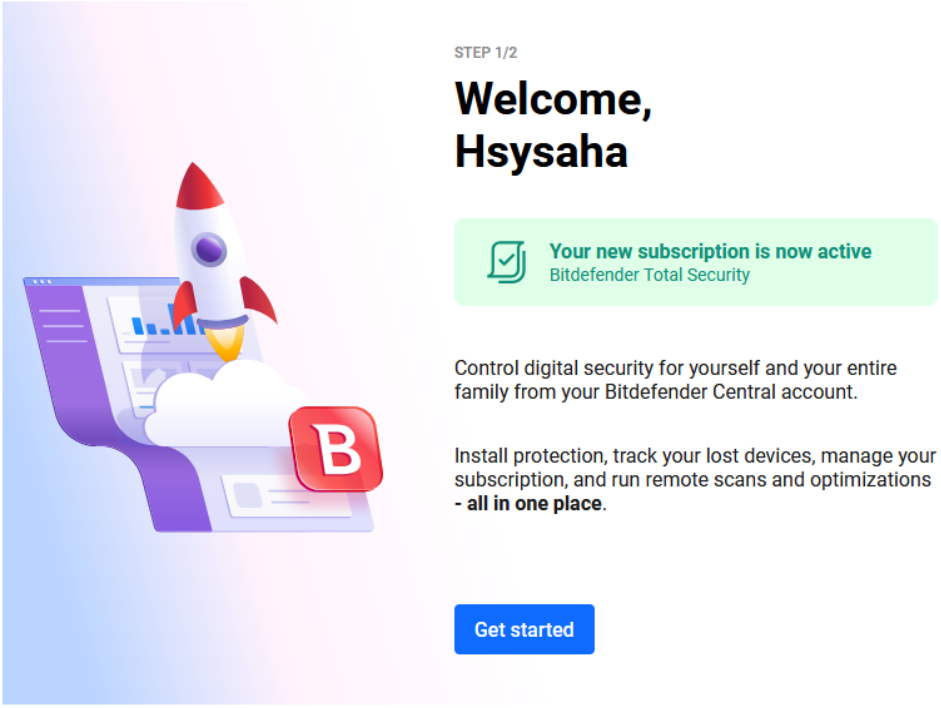
Browse to install Bitdefender (Select free for 1 month) (Server Setup)



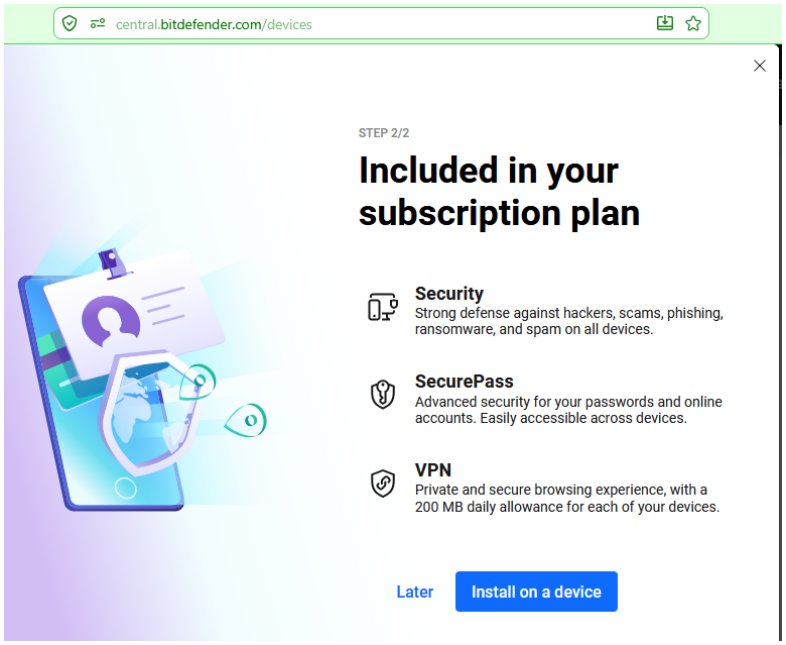
Click on "Get 1 Month FREE"



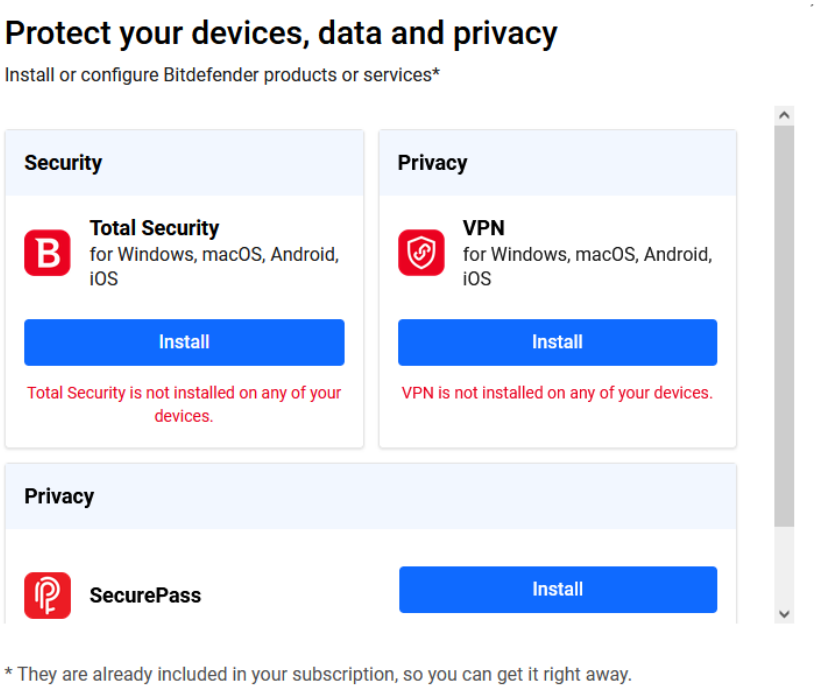
It will ask you for authentication and verifying the payment method after that sign in the bitdefender with your google account



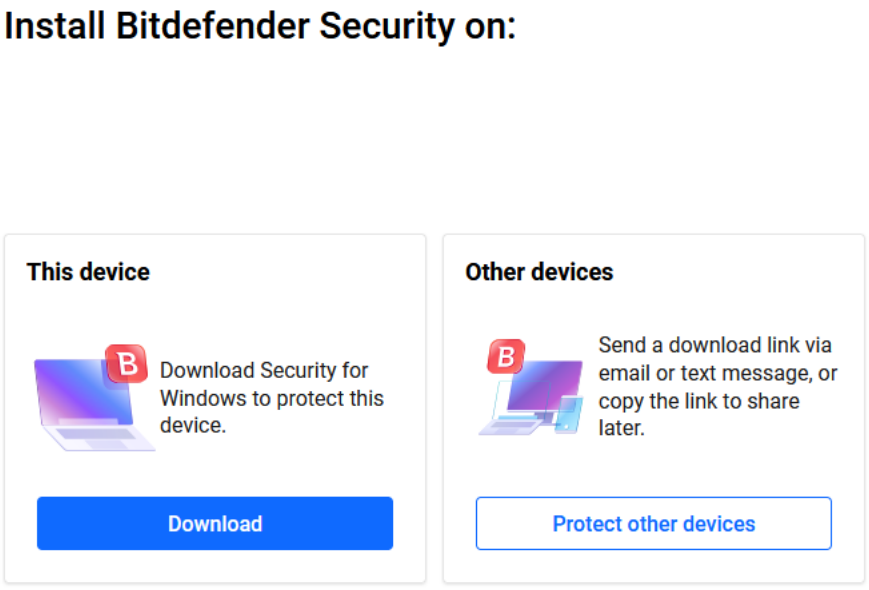
Click on installation on a device to generate the QR Code



Click on total security installation

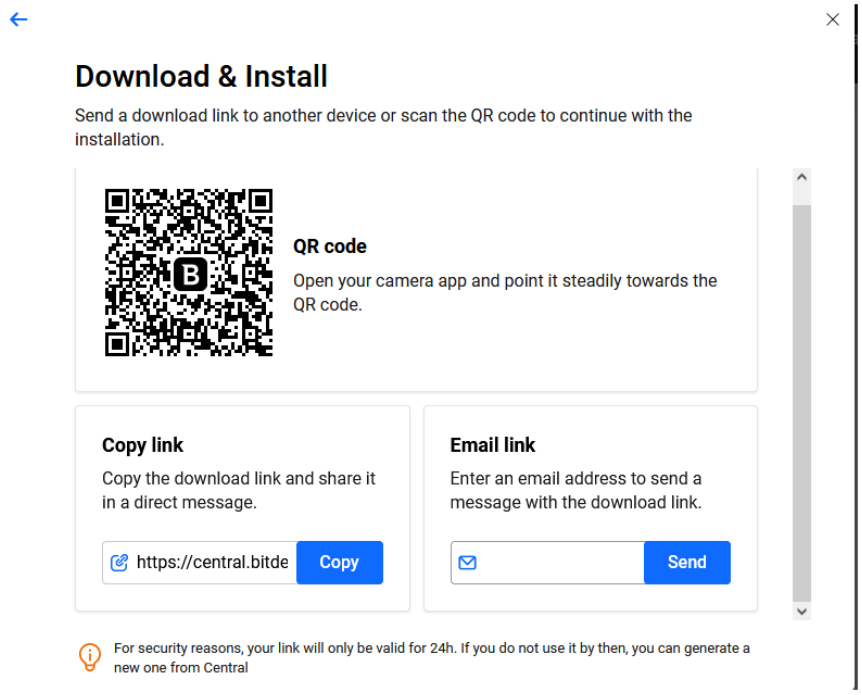


Now select protect other devices

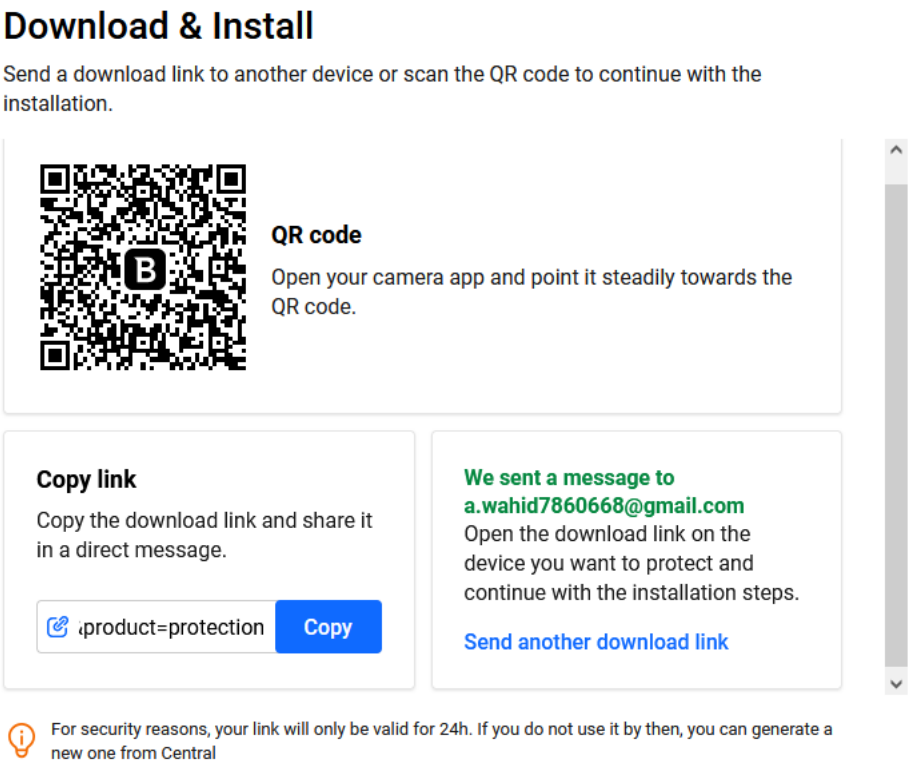


Now click on copy and paste that link in **(your phone browser)**

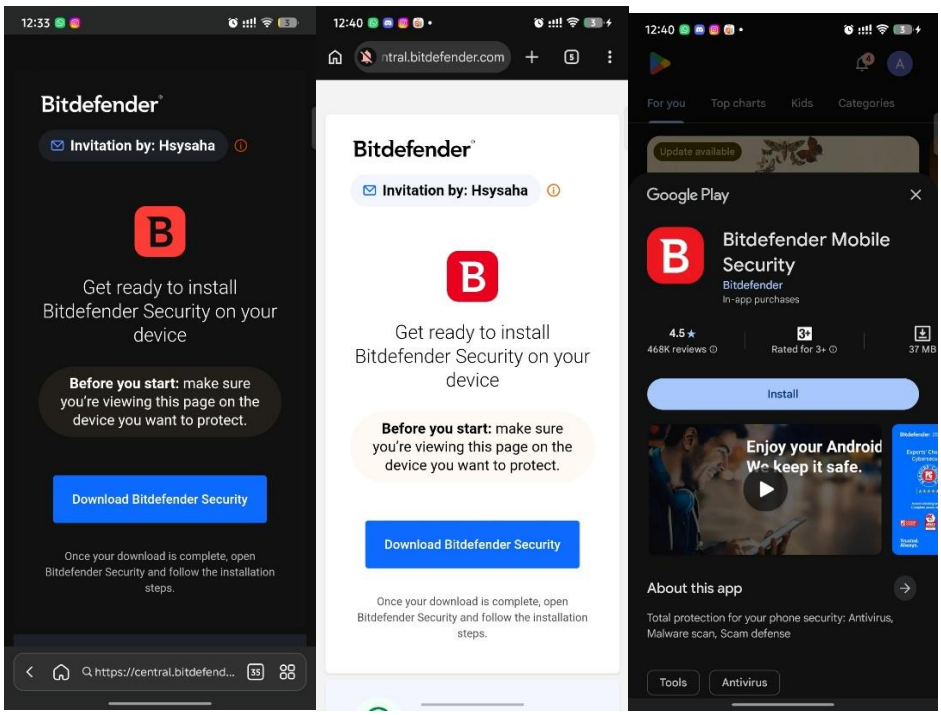
Or you can send the link via mail



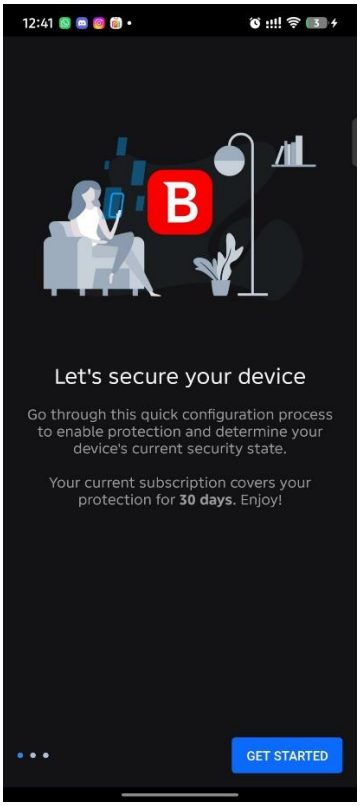
Extra (Sending link via mail)



Mobile ConfigurationTurn on your Mobile & check the Mail to Download Bitdefender in your Phone

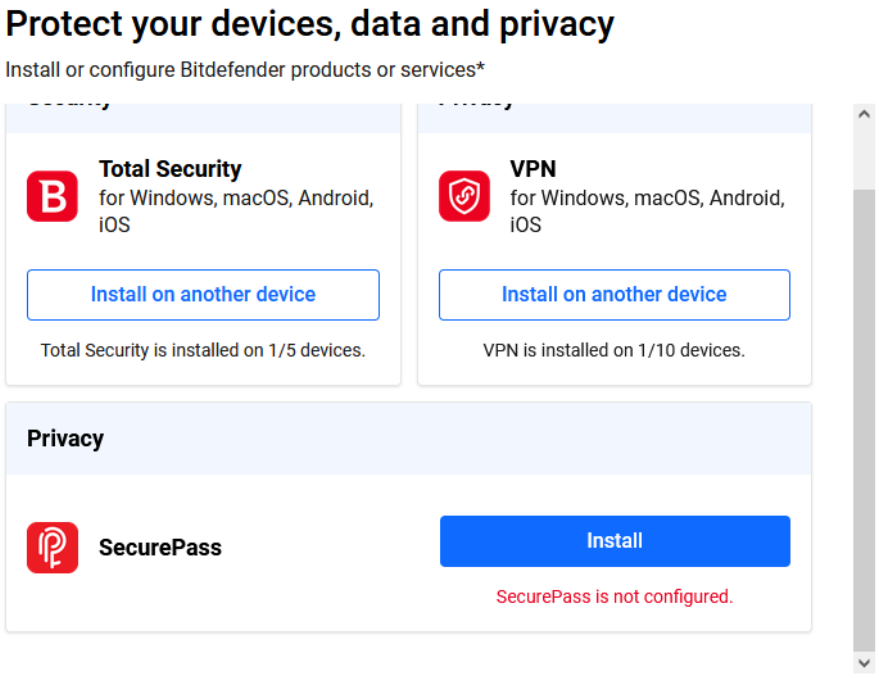


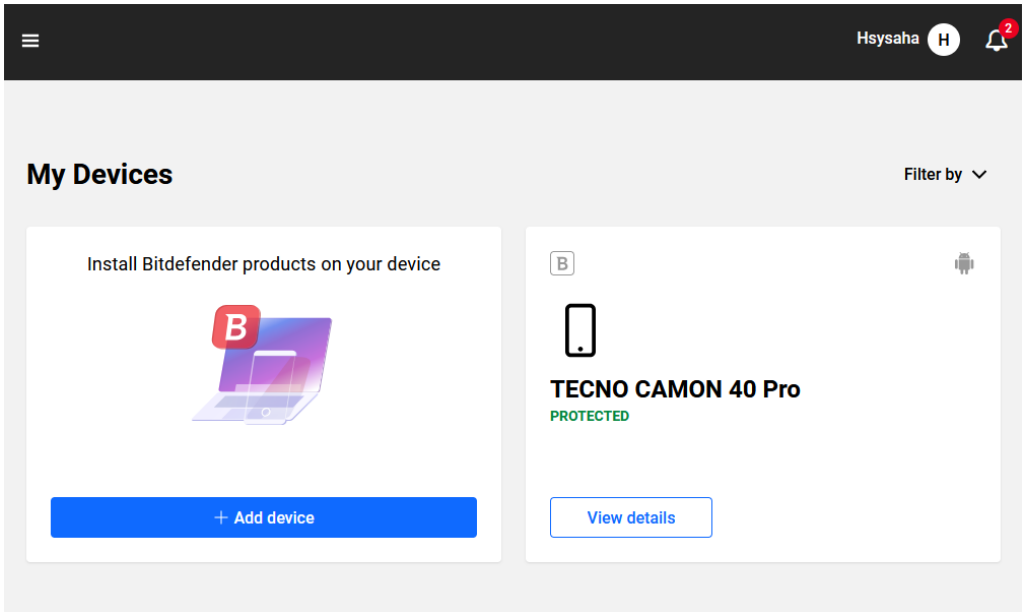
Run Bitdefender



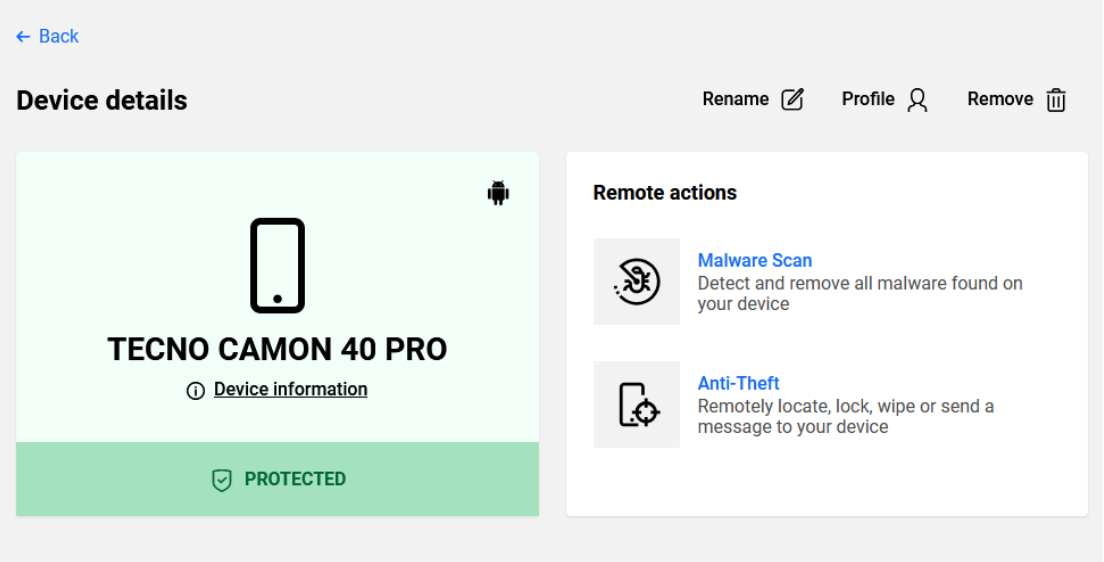
We can see our device has successfully connected to the Bitdefender

Comeback to Web server & See Your Mobile phone will be connected to Bitdefender



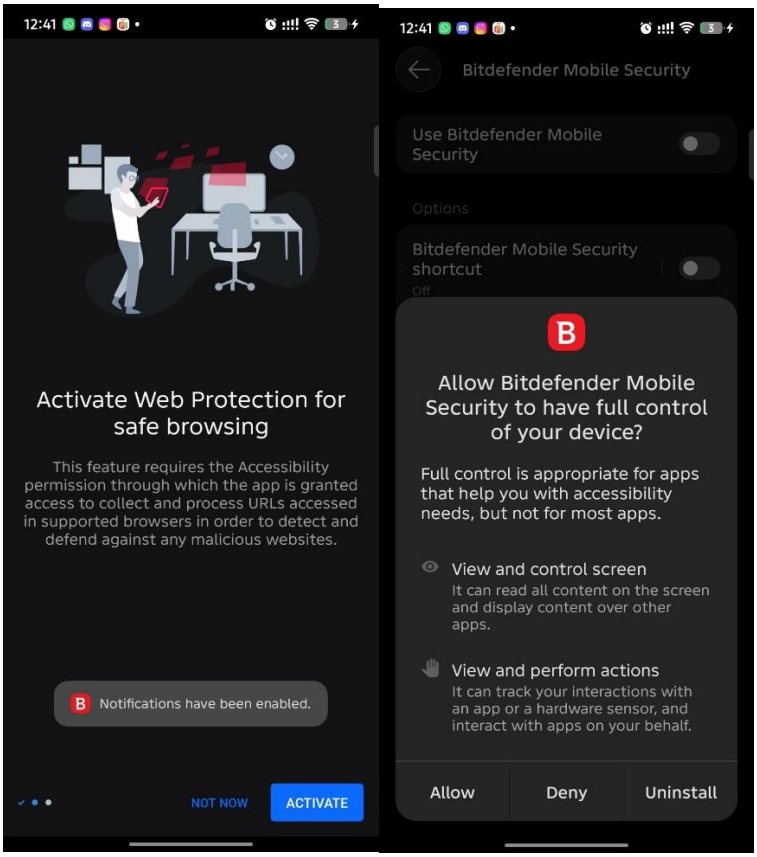


Bitdefender Connection & Also showing remote actions that can be performed from the Server

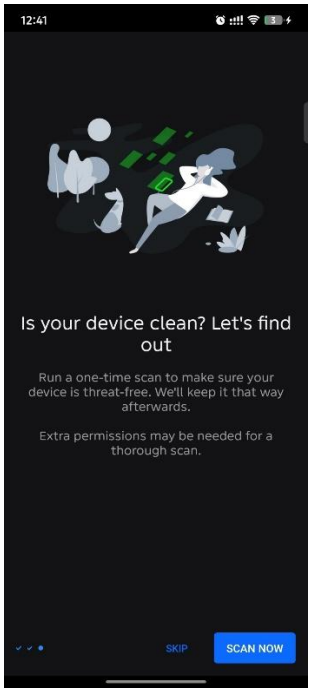


Make the Bitdefender enable all the essential resources

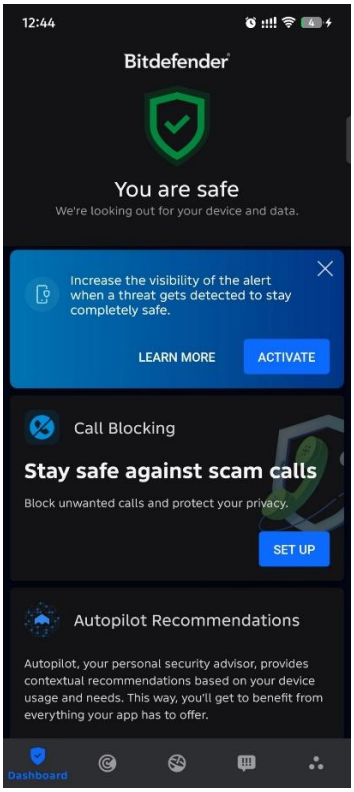
Click Activate for Web Protection



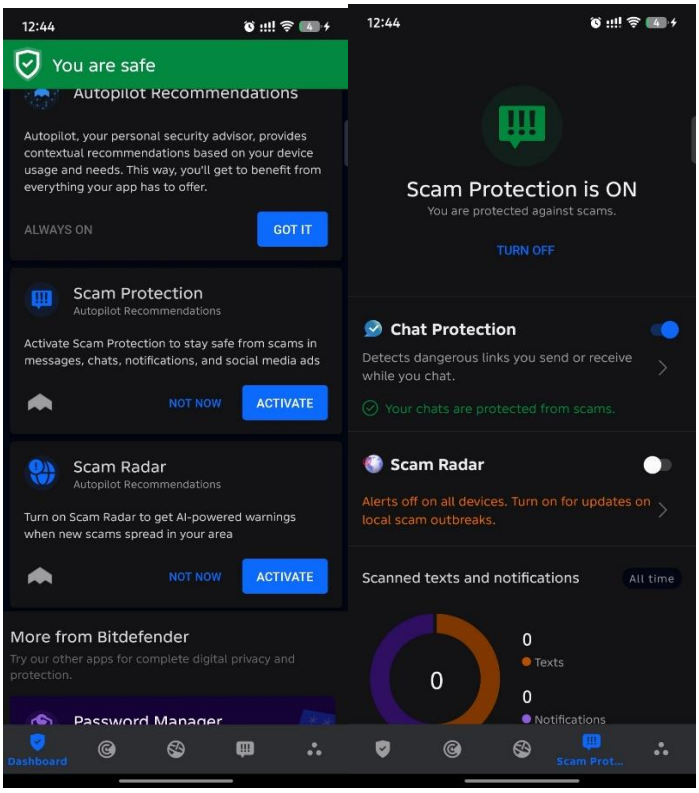
Click Skip for Now



Dashboard for Bitdefender



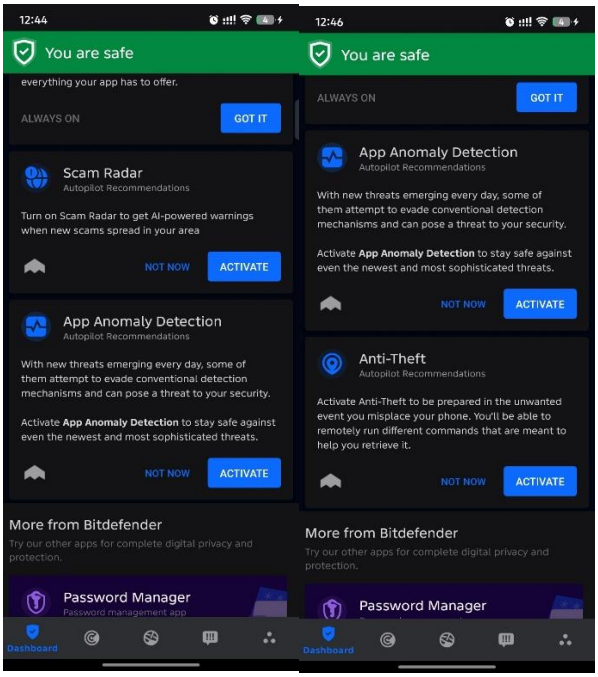
Enable Scam Protection



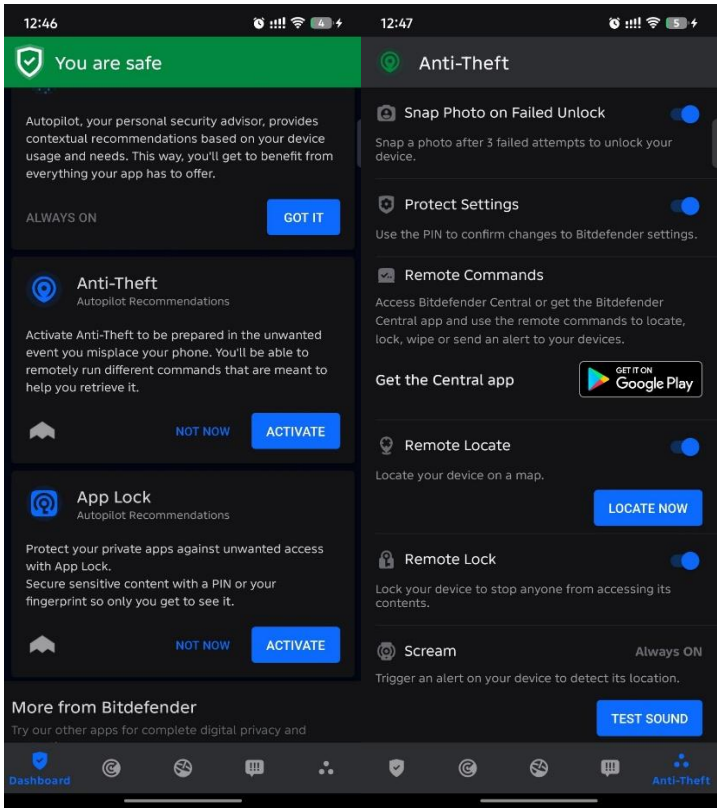
Activate

Scam radar

Anomaly Detection



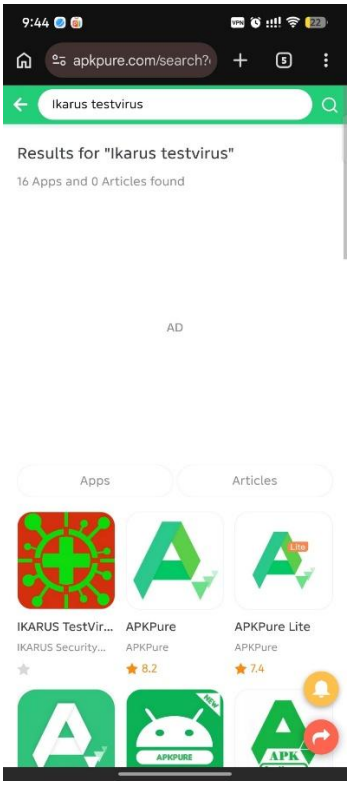
Activate Anti-Theft



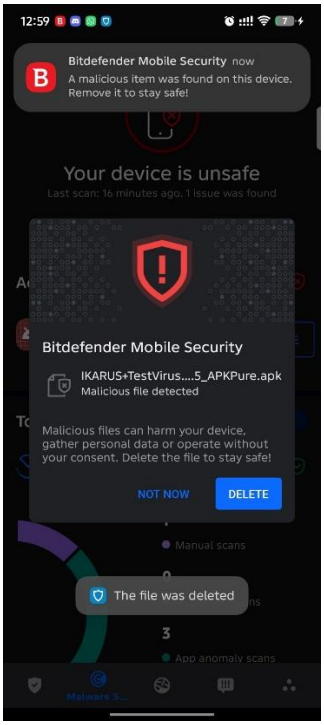
Now install the IKARUS

Detecting whether the Bitdefender checks it or not

Search & Download the IKARUS

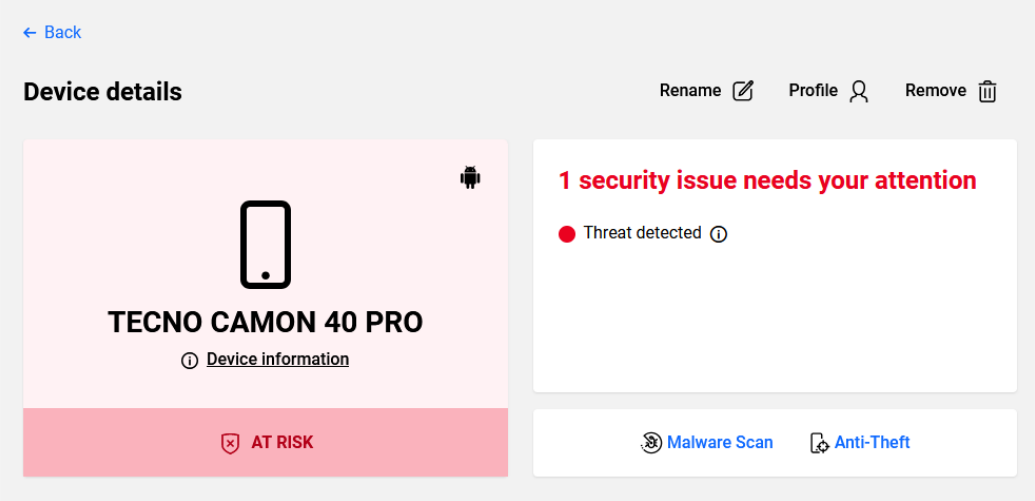


Bitdefender detected the VIRUS Successfully

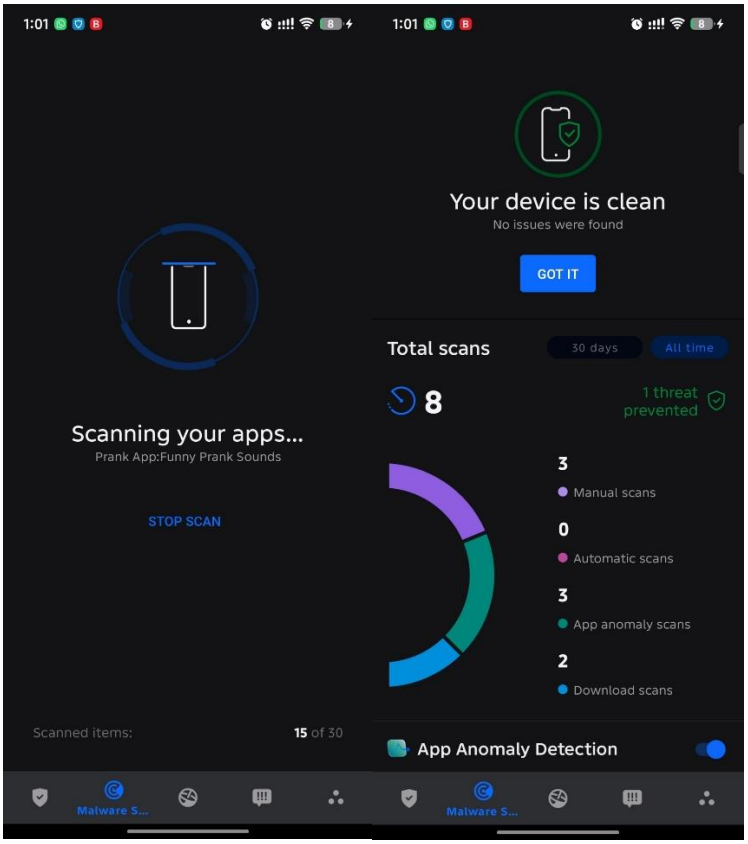


Centralized check (Server)

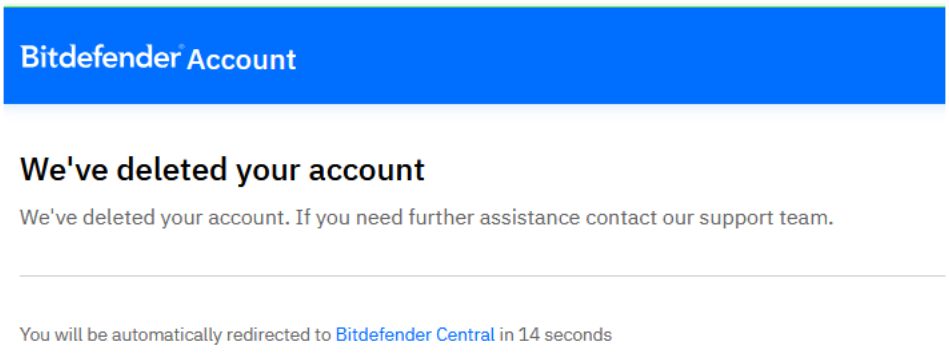
See the result on dashboard



Rescan your Phone



Delete your Bitdefender account if it's just for a lab



Conclusion

The completed labs demonstrate the importance of implementing layered security controls for mobile devices. The Emsisoft Mobile Security lab showed how standalone security tools can identify and remove malicious files through malware analysis and threat detection processes. The Bitdefender Mobile Security lab further expanded this concept by demonstrating both personal device protection and enterprise-level security management through centralized monitoring.

Together, these implementations highlight that effective mobile security requires a combination of **endpoint protection, continuous monitoring, threat detection, and centralized management capabilities**. By applying these security approaches, users and organizations can reduce mobile security risks, protect sensitive information, and respond efficiently to modern cyber threats.